Recursion and Recursive Programming

1. Read the document "The Handshake problem" by J.Spring, convert the pseudo-code on the last slide to the Python code

2. Exercise: Write a Python Function to **Find the Sum of Digits Recursively**

   Read and understand the provided explanation carefully before writing the code.

   2.1. Write the following code to define the function *sum_of_digits*:

```python
def sum_of_digits(n):
    """
    This function calculates the sum of digits in a non-negative integer n recursively.

    Args:
        n: The non-negative integer for which to calculate the sum of digits.

    Returns:
        The sum of digits in n.
    """

    # Replace the following comment with your code
    # ...
```

   2.2. Complete the function body based on the provided explanation:

   - The *if* statement should check if *n* is equal to 0.
     - If true, return 0 (base case).
   - If *n* is not 0, perform the following steps:
     - Extract the last digit using *n* % 10.
     - Make a recursive call to *sum_of_digits(n // 10)* to find the sum of digits in the remaining number.
     - Return the sum of the last digit and the result of the recursive call (n % 10 + sum_of_digits(n // 10)).

   2.3. Below the function definition, add the following code to test the function:

```python
# Example usage
number = 1234
result = sum_of_digits(number)

print(f"The sum of digits in {number} is: {result}")
```

   2.4. Run the code and observe the output. You should see the following message:

```
The sum of digits in 1234 is: 10
```

3. Exercise: Write a recursive Python program to find **the sum of the first n natural numbers.**

For example, if you enter Input: 5
Output: 15

It is calculated as:

1 + 2 + 3 + 4 + 5 = 15